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PICES science in 2019: Notes from the Science Board Chair

What can PICES do in an era of global change? The impacts of global warming and other anthropogenic perturbations became more visible and significant than ever in 2019, and they are a serious threat for the sustainability of our society. Not only warming or heat waves, but unexpected fluctuations in climate, overfishing, habitat destruction, *etc*., also damage ecosystem components and degrade ecosystem services on which our society is dependent. In December 2017, the General Assembly of the United Nation proclaimed a UN Decade of Ocean Science for Sustainable Development, 2021–2030 (hereafter, Ocean Decade). This is a clear request from society to marine scientific organizations and institutions across the globe to prepare the best scientific knowledge with appropriate timing to enable science-based decision making that will prevent further degradation of marine ecosystems and to sustain our civilization. Because global change issues are quite complex in mechanisms and components, it is essential to include stakeholders and international collaboration of both natural and social scientists in tackling these issues. PICES has a long history and experience of transdisciplinary sciences under the scheme of international collaboration. I believe that PICES will be a leading organization of global change marine science in the coming decade. We should not miss the last chance to prevent catastrophic degradation of marine ecosystem services. Under the critical situation of the Earth and society, PICES carried out various scientific activities in 2019 and established a foundation for future progress of North Pacific marine sciences.



Empress Hotel, Victoria.

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Copepod responses to, and recovery from, the recent marine heatwave in the Northeast Pacific

by Jennifer Fisher, David Kimmel, Tetjana Ross, Sonia Batten, Eric Bjorkstedt, Moira Galbraith, Kym Jacobson, Julie Keister, Akash Sastri, Karyn Suchy, Samantha Zeman and R. Ian Perry

Introduction

The anomalously warm water that formed in the North Pacific during the winter of 2013/2014, nicknamed "the Blob" (Bond *et al.,* 2015), persisted for several years and was a major and unprecedented event in the Northeast (NE) Pacific. This marine heatwave was associated with significant impacts on many marine species, disrupted typical food webs, and caused closures of several commercially important fisheries throughout the region (*e.g.*, McCabe *et al*., 2016). Copepods are ubiquitous members of marine zooplankton assemblages worldwide, and are key components of marine food webs, transferring energy from phytoplankton to fish and marine mammals. Given this ecological role and importance to fisheries, there are multiple zooplankton monitoring programs in the NE Pacific that span a large geographic region (Fig. 1).

Shifts in copepod species composition in response to the recent marine heatwave have been documented from some locations, yet the response to this extreme event at the scale of the NE Pacific has not been addressed. For example, off Newport, Oregon, the copepod community shifted to a community comprised of smaller lipid-poor taxa, greatly altering the bioenergetics of the food chain (Peterson *et al*., 2017). In the Gulf of Alaska, Batten *et al*. (2018) found that the positive relationship between temperature and zooplankton biomass from 2000 to 2013 broke down during the warm years of 2014 and 2015. They concluded such extreme warm events could alter trophic patterns and interactions. Understanding the effects of warm events is critical as they could become more prevalent and widespread (Oliver *et al*., 2018).

In this article, we examine the responses (and possible recovery) of copepods to this extreme event at the scale of the NE Pacific. We focus on two species: *Calanus pacificus*, representing a warm water copepod typically found in high abundances off California and southern Oregon but occurring throughout the region; and *Acartia longiremis*, representing a cool water species typically found in high abundances in British Columbia and Alaskan waters. We selected these two species to contrast the responses of typical warm water, and typical cold water, copepods in the NE Pacific.

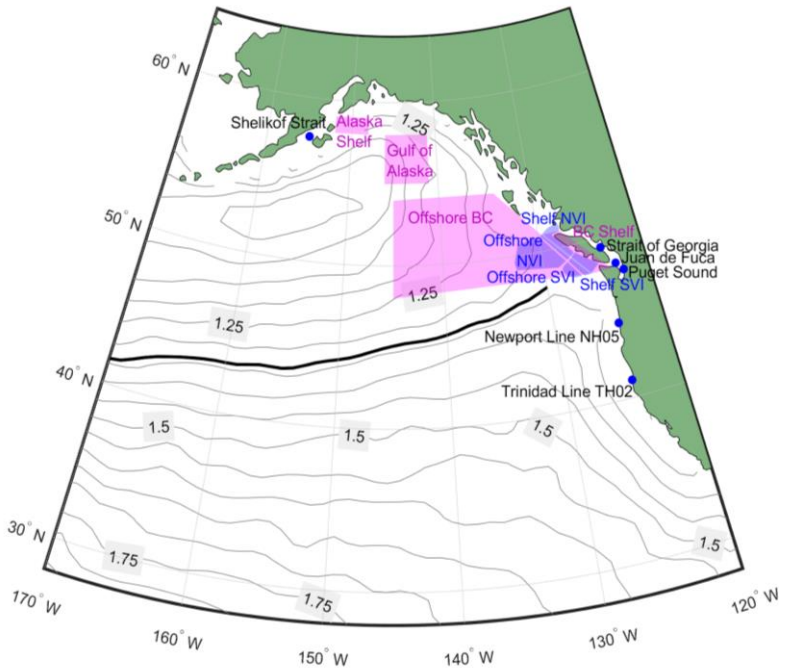


Fig. 1 Map of zooplankton monitoring locations in the NE Pacific. Contour lines show the mean dynamic height across the region based on a mapping of Argo data (2004–2019). The thick black line indicates the mean bifurcation latitude; on average the water north of the line flows north after hitting the coast of North America and the water south flows south.

Approach

We assembled data from a range of sampling programs in the NE Pacific (Table 1). Since these programs use a variety of different gears and sampling protocols, we reduced the data to Spring–Summer (May–September) anomalies for each year. Anomalies were calculated by subtracting the May–September mean abundance in each year from the long-term (May–September) mean abundance across years, and then dividing by the standard deviation of the means. The coherence in the response of the two copepod species (*C. pacificus* and *A. longiremis*) across the regions was visualized using cluster analysis. Years that are linked in the cluster dendrogram represent periods when the pattern of species abundance was more similar across regions compared to years that are not linked.

To provide the physical background within which to compare these copepod anomalies, we used Argo float data to look at the circulation of the NE Pacific. We used the Roemmich-Gilson gridded Argo temperature and salinity climatology (Roemmich and Gilson, 2009) to calculate and map the monthly mean dynamic heights of sea level (Fig. 1). We used the contours of dynamic height (the streamlines) to locate the bifurcation streamline (*i.e.*, the streamline which divides the water that will move north along the coast from that which will move south; the mean bifurcation streamline is shown as a thick black line in Fig. 1). Using the dynamic height of the bifurcation, we then calculated the differences in dynamic height between the bifurcation and a point in the middle of the Alaskan gyre (53°N, 155°W; ΔD_{Gak}) and a point off the coast of California (35°N, 150°W; ΔD_{CCurr}). These differences in dynamic height then scale with the flow going north (ΔD_{Gak}) and south (ΔD_{CCurr}) along the coast of North America (Fig. 2). Additionally, we calculated the difference between the locations off the coast of California and the Gulf of Alaska (ΔD_{NPC}), which scales with the overall magnitude of the North Pacific Current (NPC) in the NE Pacific (Fig. 2). Thus, this analysis provides information on the mean latitude of the center of the NPC (shown to be important in Sydeman *et al*., 2011), *i.e.*, the mean latitude of flow across the NE Pacific, and the fraction of that flow that moves south along the Oregon and California coasts (f_{CC} in Fig. 2) or north into the Gulf of Alaska ($1\text{-}f_{CC}$).

Table 1 Zooplankton sampling programs in the NE Pacific used in this analysis. See Figure 1 for locations.

Program	Location	Provider	Year range
Trinidad Head Line	Northern California	Eric Bjorkstedt	2008–2016
Newport Hydrographic Line	Central Oregon	Jennifer Fisher, Kym Jacobson, Samantha Zeman	1996–present
Puget Sound	Puget Sound	Julie Keister	2014–present
Strait of Juan de Fuca	Strait of Juan de Fuca	Julie Keister	2003–2017
Strait of Georgia	Strait of Georgia, BC	Moira Galbraith, Ian Perry, Kelly Young, Karyn Suchy	1996–2018
La Perouse/West Coast of Vancouver Island	Southern continental shelf off Vancouver Island; Southern off-shelf waters of Vancouver Island; Northern continental shelf off Vancouver Island; Northern off-shelf waters of Vancouver Island	Moira Galbraith, Ian Perry, Akash Sastri	1990–2019
Shelikof Strait	Shelikof Strait, Alaska	David Kimmel	1990–present
Continuous Plankton Recorder program – Pacific (CPR-Pacific)	Four regions were delineated, two offshore (northern Gulf of Alaska and west of BC) and two shelf regions (Central Alaska and BC)	Sonia Batten	2000–present (offshore) 2004–present (shelf)

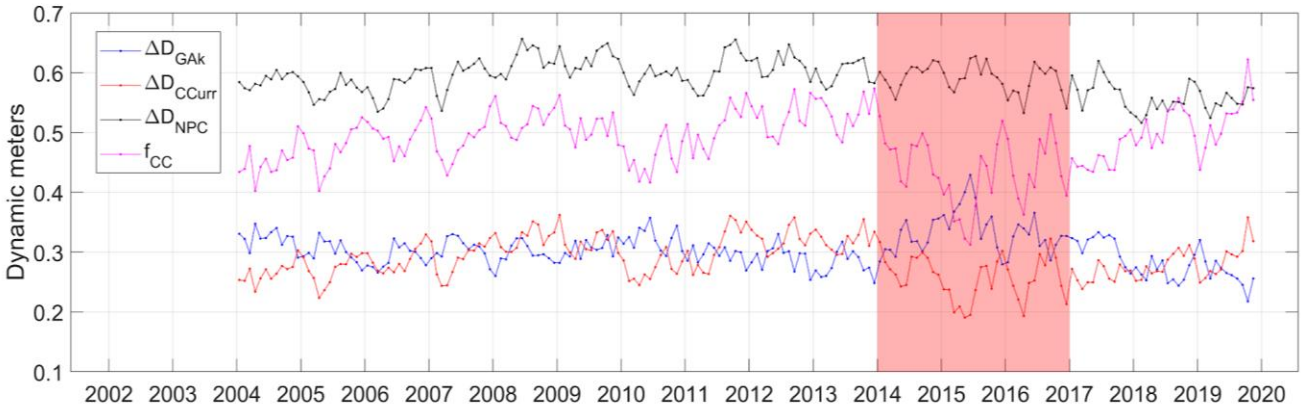


Fig. 2 Time series of differences in dynamic heights (in m) between the bifurcation streamline for the North Pacific Current (NPC) and the Gulf of Alaska (ΔDGak in blue), the bifurcation streamline and a point off the California coast (ΔDCCurr in red) and across the NPC (ΔDNPC in black). The pink line (fCC) shows the ratio ΔDCCurr/ΔDNPC, which is the fraction of the NPC that flows south along the coast of North America. The shading highlights the recent marine heatwave years in the open NE Pacific (2014–2017).

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What we found

Strong changes in the fraction of water transported from the NPC into the California Current began in 2014 and persisted through 2017 (Fig. 2). Note the lack of a large change in the overall strength of the NPC during this time, so the flow into the Gulf of Alaska became stronger and the flow into the California Current became weaker.

At the onset of the unusually high fraction of the NPC turning north in 2014, positive anomalies of *C. pacificus* started occurring at many of the northern locations in the NE Pacific (Fig. 3; Alaska shelf, Gulf of Alaska offshore, South Vancouver Island offshore and on the shelf, Strait of Georgia and Puget Sound). However, they remained negative in open coast locations to the south (Fig. 3; Newport and Trinidad lines). These patterns follow the temporal evolution of the marine heatwave that started in the Gulf of Alaska and offshore in the NE Pacific, but did not inundate continental shelf waters at lower latitudes until coastal upwelling winds subsided in the fall of 2014. From 2015–2017, positive anomalies of *C. pacificus* occurred across almost all of the regions while negative (or neutral) anomalies of *A. longiremis* occurred across all regions (Fig. 4).

The coherence of abundance changes of these two species across the regions was stronger for *C. pacificus* compared to *A. longiremis*. Years associated with the marine heatwave (2015–2017) clustered together with 2005, another warm ocean year (Mackas *et al*., 2006), indicating that the shifts in abundance of *C. pacificus* across regions were similar during these years (Fig. 5A). The grouping of the warm years for *A. longiremis* is less consistent, with 2016 and 2005 clustering together, and 2015 and 2017 clustering with 2004 (Fig. 5B).

Weak, but negative anomalies of *C. pacificus* abundance occurred in many regions in 2018 and the coherence of the signal across the regions meant that this year clustered with other years in the study, indicating that the copepod community might have been stabilizing in 2018. However, strongly positive anomalies of *C. pacificus* occurred on the shelf off North and South Vancouver Island in 2019 and *A. longiremis* abundance remains below average across many of the regions. It appears the effects of the marine heatwave remain apparent in the copepod community despite a decline in anomalously high ocean temperatures.

Implications

Despite variability in the abundance of *C. pacificus* and *A. longiremis* across these monitoring regions throughout each time series, patterns in response to the recent marine heatwave emerged. Positive anomalies of *C. pacificus* and negative anomalies of *A. longiremis* generally occurred across all of the regions, as would be expected given the spatial extent and magnitude of the positive temperature anomalies and the water mass affinities of these two copepod species. The coherence of abundance changes of these two species across regions was also similar in 2005, however, the changes in abundance of the two species was not. During 2005, positive anomalies of both species occurred in the northern regions while mostly negative anomalies of both species occurred across the central to southern regions. Positive temperature anomalies also occurred during 2005, however, the mechanism, magnitude, and geographic extent of this warm anomaly was different compared to the marine heatwave. Therefore, when assessing the possible impact a physical perturbation might have on lower trophic level communities, it is important to understand the magnitude and geographic extent of the event in question.

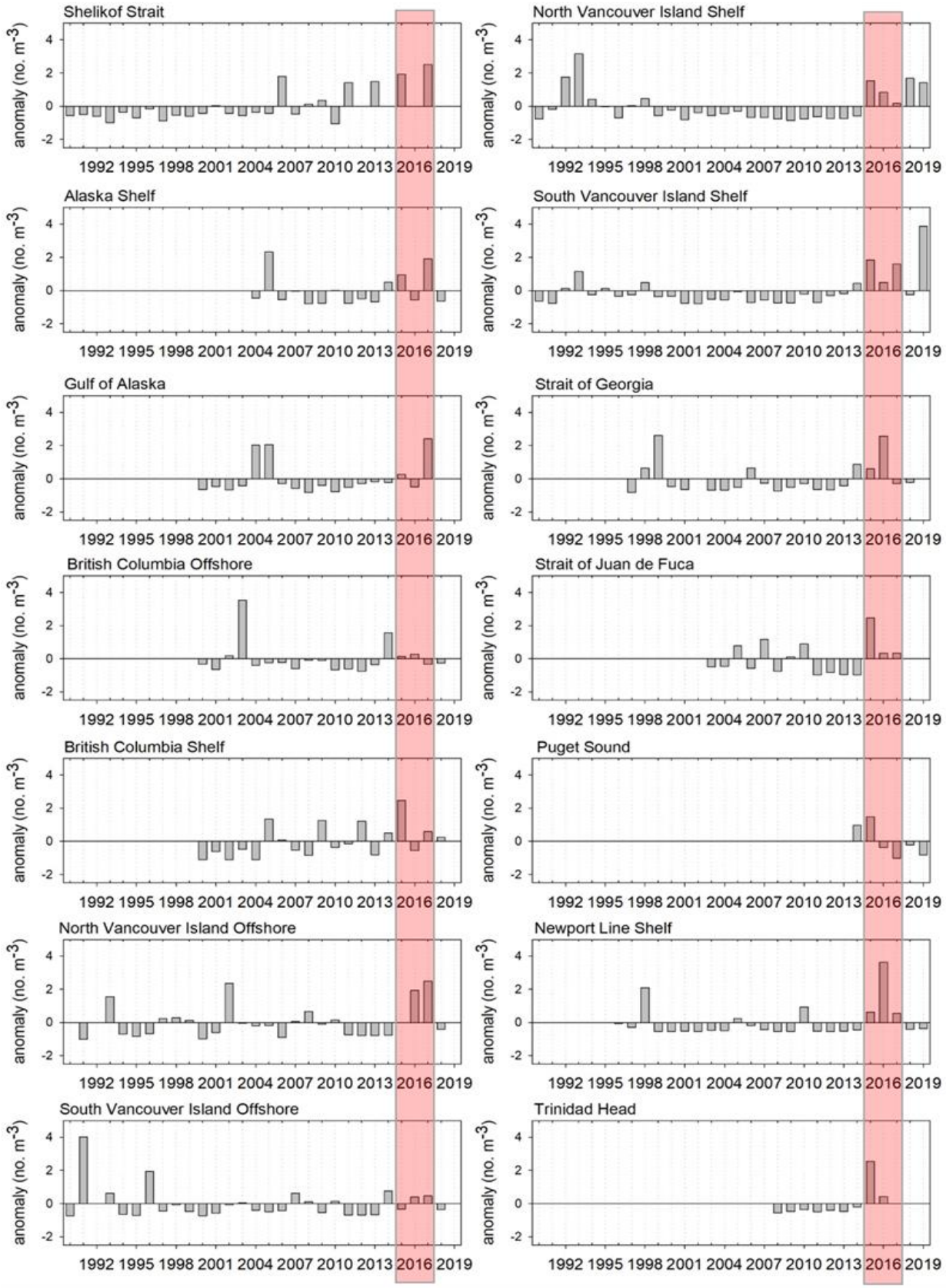


Fig. 3 Annual anomalies of Calanus pacificus density (no. m^{-3}) from May–September across the study regions. Shaded regions represent the recent marine heatwaves years along continental and adjacent waters of the NE Pacific (2015–2017).

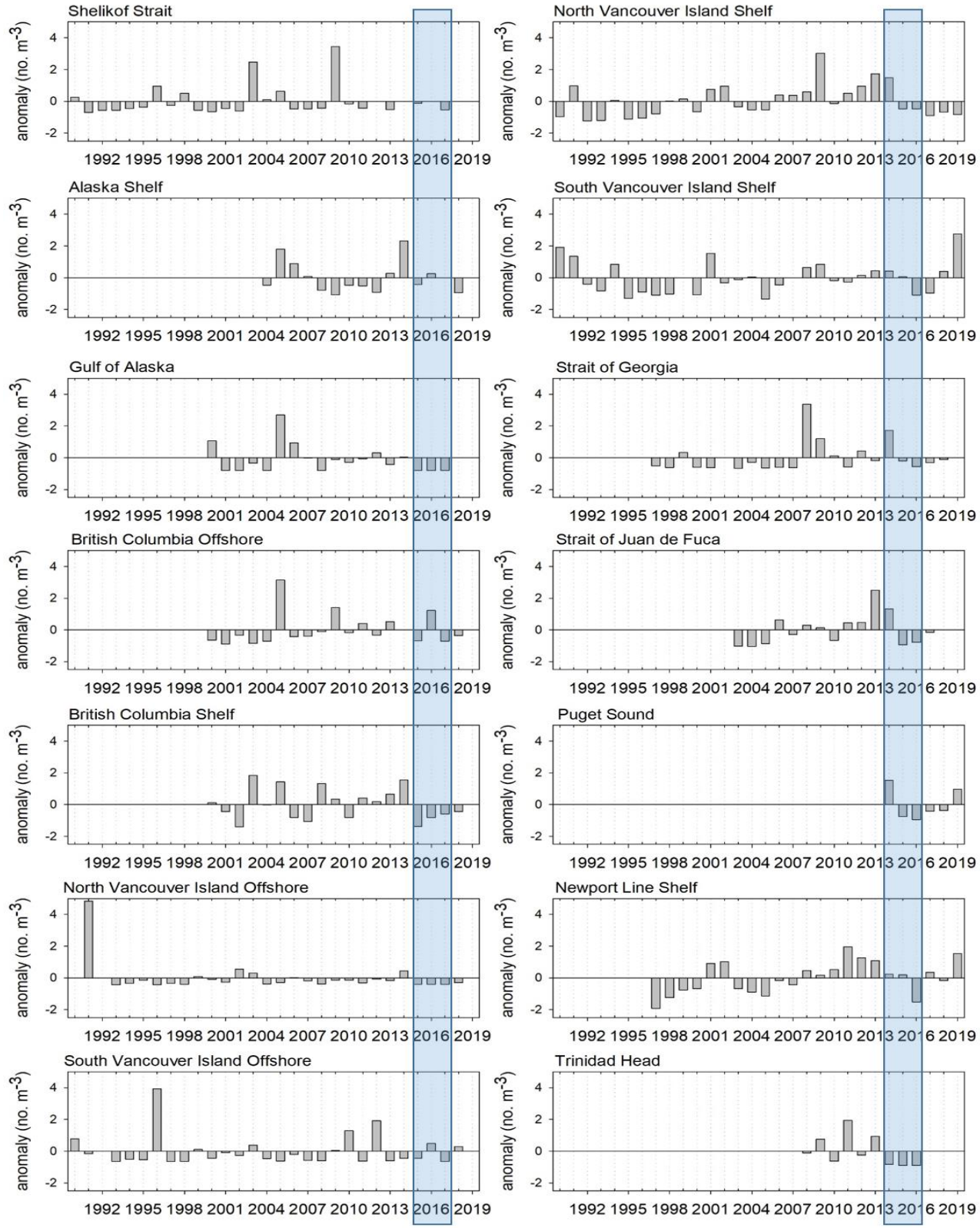


Fig. 4 Annual anomalies of Acartia longiremis density (no. m^{-3}) from May–September across the study regions. Shaded regions represent the recent marine heatwaves years along continental and adjacent waters of the NE Pacific (2015–2017).

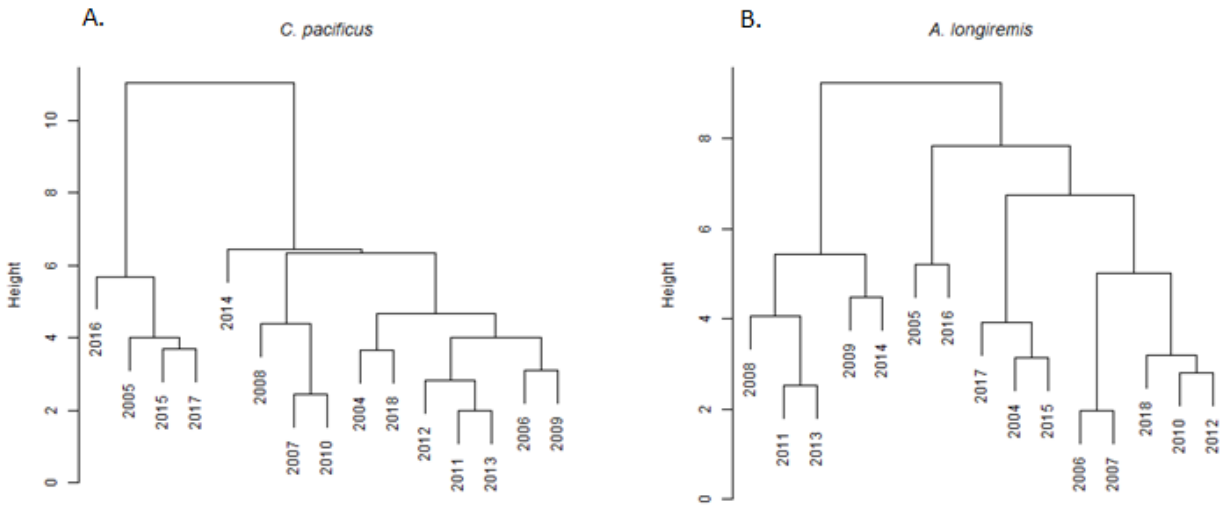


Fig. 5 Cluster dendrogram showing the similarity in abundance shifts for A. Calanus pacificus and B. Acartia longiremis across the regions from 2004–2018. Dissimilarity across regions was calculated using Euclidean distance on square root transformed densities for each species.

The marine heat wave has resulted in a variety of biological impacts from reduced chlorophyll biomass to species range shifts and bird die-offs. From the copepod perspective, the positive anomalies of *C. pacificus* throughout the region may have consequences for trophic transfer in NE Pacific food webs. This species is smaller and has lower lipid storage reserves compared to its congener, *C. marshallae*. A prey base with a greater proportion of *C. pacificus* relative to *C. marshallae* would be most impactful in northern latitudes where the latter is an important resource supporting juvenile fish, adult forage fish, and bird populations.

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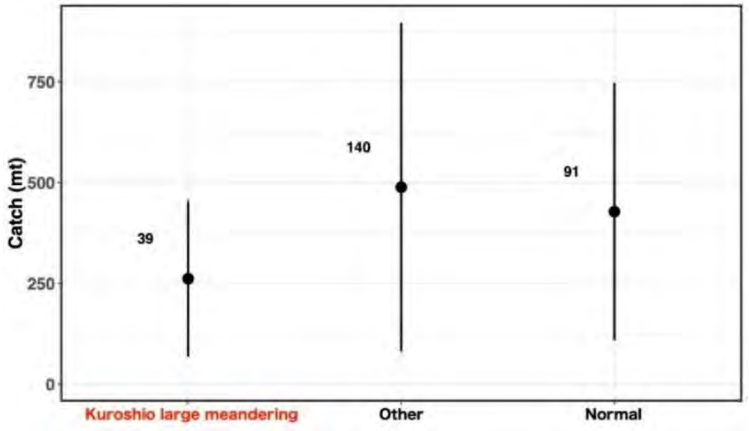


Fig. 4 Skipjack catches (mt) in Wakayama Prefecture from coastal trolling activities based on Kuroshio current trajectories. Large Kuroshio meanders are defined as those with a latitude of 32°N between 136°E and 140°E, while the normal situation is defined as straight flow along Japan's southern coast and other instances are defined as neither of the above. Numbers next to bar plots represent monthly data counts



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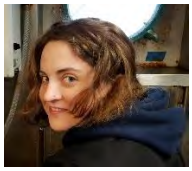
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